

Cosmology Models, Dark Energy & 21 cm Hydrogen — Teaching Notes

CASSA RA 2026-27 · the "theory" companion

Goal: understand the models, not just memorize them. Each idea has a plain explanation + a small worked example. Read slowly once; the math is deliberately easy.

PART 1 — How we model the whole Universe

1.1 The one big assumption

On large scales the Universe looks the **same everywhere (homogeneous)** and the **same in every direction (isotropic)**. This is the **Cosmological Principle**. It lets us describe all of space with a single number that grows with time: the **scale factor $a(t)$** .

- $a(\text{now}) = 1$. In the past $a < 1$, in the future $a > 1$.
- Redshift links to it: $1 + z = 1/a$.
- *Example:* light from $z = 3$ was emitted when $a = 1/(1+3) = 0.25$, i.e. the Universe was **1/4 of its present size**.

1.1b Types of redshift (and which frame you measure in)

"Redshift" $z = (\lambda_{\text{obs}} - \lambda_{\text{emit}})/\lambda_{\text{emit}}$ can come from **three different physical causes**. In supernova cosmology you must separate them, because only one carries the cosmology.

Three physical origins: 1. **Cosmological redshift** — the **expansion of space** stretches the wavelength while light travels. This is $1+z = 1/a$. It is *not* a real Doppler shift; nothing is "moving through" space. This is the redshift that maps to distance on the Hubble diagram. 2. **Doppler (peculiar-velocity) redshift** — a galaxy's own **motion through space** (falling toward a cluster, etc.), on top of the Hubble flow. Radial formula: $1+z = \sqrt{[(1+\beta)/(1-\beta)]}$, $\beta = v/c$; for small v , $z \approx v/c$. - *Example:* a peculiar velocity of 300 km/s gives $z \approx 300/300000 = 0.001$. Tiny, but at low redshift ($z \approx 0.02$) that is a **5% error** in z , so it must be corrected. 3. **Gravitational redshift** — light **climbing out of a gravitational well** loses energy: $1+z \approx 1 + GM/(rc^2)$. A general-relativity effect (also gives the Sun's ~ 0.6 km/s line shift and clocks on the CMB).

They multiply: $(1 + z_{\text{obs}}) = (1 + z_{\text{cosmo}})(1 + z_{\text{pec}})(1 + z_{\text{grav}})$.

Reference frames (the "helio vs CMB" part) — a measurement convention, not a new physics: - **Topocentric** — raw redshift as seen from the moving telescope on the spinning, orbiting Earth. - **Heliocentric (or barycentric) redshift z_{helio}** — corrected for **Earth's motion** (spin ~ 0.5 km/s + orbit ~ 30 km/s) so it is referred to the **Sun**. This is the standard catalogue value. - **CMB-frame redshift z_{CMB}** — further corrected for the **Sun's own motion relative to the cosmic microwave background**. We move at **~ 370 km/s toward the constellation Leo**, which we see as the **CMB dipole** ($z_{\text{dipole}} \approx 370/300000 \approx 0.0012$). The CMB frame is the closest thing to the **cosmic rest frame**, so cosmology is done here. - **Why it matters for your SN work:** the Hubble diagram needs z_{CMB} , and at low redshift you also **correct peculiar velocities** (from local flow models), because they are the dominant scatter there. So the pipeline goes: *observed* $\rightarrow$ *heliocentric* $\rightarrow$ *CMB frame* $\rightarrow$ *peculiar-velocity corrected*. Good line for the viva.

1.2 The metric (FLRW)

The geometry is the **Friedmann-Lemaitre-Robertson-Walker (FLRW)** metric. You do not need the full tensor form; the idea is: distances between galaxies scale as $a(t)$. Galaxies are like dots on an inflating balloon, the balloon surface is a .

1.3 The Friedmann equations (the engine)

From Einstein's equations with the FLRW metric you get two equations.

(A) Expansion rate: $H^2 = (\dot{a}/a)^2 = (8\pi G/3) \rho - kc^2/a^2 + \Lambda c^2/3$

- H = Hubble parameter (expansion rate), ρ = total energy density, k = curvature (+1,0,-1), Λ = cosmological constant.

(B) Acceleration: $\ddot{a}/a = -(4\pi G/3)(\rho + 3P/c^2) + \Lambda c^2/3$

- Key physics: **pressure gravitates too**. Normal matter ($P \approx 0$) slows expansion. But something with **large negative pressure** can make $\ddot{a} > 0 \rightarrow$ **acceleration**. That is dark energy.

1.4 Critical density and Ω

Define the **critical density** (the density for a flat universe): $\rho_{\text{crit}} = 3H^2 / (8\pi G) \approx 9 \times 10^{-27} \text{ kg/m}^3$ today (about 5 hydrogen atoms per cubic metre).

Then each component is a fraction $\Omega = \rho / \rho_{\text{crit}}$: $\Omega_{\text{matter}} + \Omega_{\text{radiation}} + \Omega_{\Lambda} + \Omega_{\text{curvature}} = 1$
Measured today (concordance): $\Omega_m \approx 0.3$, $\Omega_{\Lambda} \approx 0.7$, $\Omega_r \approx 10^{-4}$, $\Omega_k \approx 0$ (flat).

1.5 How each ingredient dilutes as space expands

General rule for a component with equation of state $w = P/(\rho c^2)$: $\rho \propto a^{-(3(1+w))}$

Component	w	Scaling	Why
Matter (dust, dark matter)	0	$\rho \propto a^{-3}$	volume grows as a^3 , mass fixed
Radiation (photons)	1/3	$\rho \propto a^{-4}$	volume a^3 plus each photon redshifts (loses energy $\propto a^{-1}$)
Cosmological constant / Λ	-1	$\rho \propto a^0 =$ constant	vacuum energy does not dilute

- *Consequence:* early Universe = radiation-dominated, then matter-dominated, now Λ -dominated. Λ "wins" late because everything else dilutes and it does not.

1.6 Friedmann equation you actually use (flat Λ CDM)

$H(z) = H_0 \cdot \sqrt{[\Omega_m (1+z)^3 + \Omega_r (1+z)^4 + \Omega_{\Lambda}]}$

- *Example (ignore radiation):* at $z = 1$, with $\Omega_m=0.3$, $\Omega_{\Lambda}=0.7$: $H = H_0 \sqrt{(0.3 \cdot 8 + 0.7)} = H_0 \sqrt{3.1} \approx 1.76 H_0$. So the Universe was expanding **~76% faster** at $z=1$ than now.

PART 2 — The standard model and its cousins

2.1 Λ CDM (the concordance model)

Λ (cosmological constant / dark energy) + **CDM** (Cold Dark Matter) + flat geometry. - Cold = dark matter moves slowly (non-relativistic), clumps to form structure. - Fits CMB (Planck), galaxy clustering/BAO, and SN Ia Hubble diagram together $\rightarrow$ "concordance".

2.2 Simple limiting models (good to name)

- **Einstein-de Sitter:** flat, matter only ($\Omega_m=1$). Gives $a \propto t^{(2/3)}$. The old pre-1998 default; decelerating forever.
- **de Sitter:** empty except Λ . Gives **exponential** growth $a \propto e^{(Ht)}$. Describes the far future of Λ CDM and (roughly) inflation.
- **Radiation era:** $a \propto t^{(1/2)}$.

2.3 The deceleration parameter (one-number test)

$q_0 = \Omega_m/2 - \Omega_{\Lambda}$ (for flat matter+ Λ). - *Example:* $q_0 = 0.3/2 - 0.7 = 0.15 - 0.7 = -0.55$. **Negative $\rightarrow$ accelerating**. This single number is what the 1998 SN Ia data flipped from positive to negative.

PART 3 — Dark energy meets theoretical physics

3.1 What could Λ be? (vacuum energy)

Quantum field theory says empty space has a **vacuum (zero-point) energy** with pressure $P = -\rho c^2$, i.e. exactly $w = -1$. That is a perfect candidate for Λ .

3.2 The cosmological constant problem (the famous embarrassment)

If you estimate the vacuum energy from QFT, you get a value about **10^{120} times larger** than what we observe. This mismatch (often called "the worst prediction in physics") is *why* dark energy is a deep theoretical problem, not just an astronomy number. Good line for a viva.

3.3 Dynamical dark energy (quintessence)

Maybe dark energy is not constant but a **slowly rolling scalar field ϕ** (like a very light "cosmic" version of the inflaton or Higgs idea). Then w can differ from -1 and **change with time**. - Parametrized often as **CPL**: $w(a) = w_0 + w_a(1 - a)$. - w_0 = value today, w_a = how it evolves. Λ CDM is the special case $w_0 = -1$, $w_a = 0$. - **DESI 2024** (BAO + SN) hinted at $w_0 > -1$ and $w_a < 0$, i.e. **possibly evolving** dark energy. *This is exactly why host-galaxy systematics like yours matter: if SN Ia carry a hidden age bias, it could fake such a signal.*

3.4 Modified gravity (the other option)

Instead of new energy, maybe **General Relativity itself is modified** on cosmic scales (f(R) gravity, DGP, etc.). Then "dark energy" is a symptom of the wrong gravity law. Distinguishing this from Λ needs growth-of-structure tests, not just distances.

3.5 The Hubble tension (live controversy)

- **Early-universe route** (Planck CMB, assuming Λ CDM): $H_0 \approx 67$ km/s/Mpc.
- **Late-universe route** (SN Ia distance ladder, SH0ES): $H_0 \approx 73$.
- The $\sim 5\sigma$ gap may mean new physics (early dark energy?) or an unknown systematic. Your field (SN standardization + host systematics) sits right at the centre of this debate.

PART 4 — 21 cm cosmology & hydrogen (your radio bridge)

4.1 Why hydrogen rules the Universe

$\sim 75\%$ of normal (baryonic) matter is **hydrogen**. Its light is how we read cosmic history. Terminology: - **HI** = neutral atomic hydrogen. **HII** = ionized hydrogen (proton). **H₂** = molecular. - **Recombination ($z \approx 1100$)**: Universe cools enough for protons + electrons $\rightarrow$ neutral H. Photons stop scattering $\rightarrow$ released as the **CMB** (the "surface of last scattering"). - **Dark Ages**: after recombination, before first stars. Only neutral H, no light sources. - **Cosmic Dawn**: first stars/galaxies form. - **Epoch of Reionization (EoR, $z \approx 6-10$)**: UV from first stars/quasars **re-ionizes** the hydrogen back to HII.

4.2 The 21 cm line (the star of the show)

Neutral hydrogen's ground state has a tiny **hyperfine splitting**: the electron and proton spins can be **parallel** (slightly higher energy) or **antiparallel** (lower). - The spin flip emits a photon of **wavelength $\lambda = 21.1$ cm, frequency $\nu = 1420.4$ MHz** (radio!). - It is a **"forbidden" transition** (a single atom flips on average once in ~ 10 million years), but there is *so much* hydrogen in the Universe that the line is easily seen. - **Worked example (why we need low-frequency radio)**: the line is redshifted. Observed frequency: $\nu_{\text{obs}} = 1420.4 / (1+z)$ MHz. - At $z = 6$ (end of reionization): $\nu_{\text{obs}} = 1420.4/7 \approx 203$ MHz. - At $z = 17$ (cosmic dawn): $\nu_{\text{obs}} = 1420.4/18 \approx 79$ MHz. - That is why 21 cm experiments are **low-frequency radio** (tens to hundreds of MHz).

4.3 Spin temperature and brightness (the key concepts)

- **Spin temperature T_s** = a temperature that encodes the ratio of atoms in the two hyperfine levels.
- What we measure is the **21 cm signal seen against the CMB**:

- If $T_s > T_{\text{CMB}} \rightarrow$ **emission** (bright).
- If $T_s < T_{\text{CMB}} \rightarrow$ **absorption** (dark dip).
- **Wouthuysen-Field effect: Lyman-alpha photons** from the first stars couple T_s to the **gas kinetic temperature** (which is colder than the CMB after cosmic dawn) $\rightarrow$ produces a 21 cm **absorption trough**. This is the mechanism behind the signal.
- **EDGES (2018)** reported an absorption dip near **78 MHz ($z \approx 17$)** that was surprisingly **deep**, hinting at either colder gas (exotic dark-matter cooling?) or an extra radio background. Still debated.

4.4 Why 21 cm is a dream cosmological probe

- It is a **3D map**: each observed frequency = a different redshift slice. Optical surveys mostly see galaxies; 21 cm sees the **hydrogen everywhere**, including the Dark Ages no other probe reaches.
- Instruments (all low-frequency radio, ties to your I-HOW workshop): **LOFAR, MWA, HERA, and future SKA-Low**.

4.5 Other hydrogen lines you should know (not 21 cm)

These are **electronic** transitions (electron changing orbit n), totally different from the hyperfine 21 cm line: - **Lyman series** (down to $n=1$, ultraviolet): **Lyman-alpha** = **1216 Å** ($n=2 \rightarrow 1$). The **Lyman limit** = **912 Å** ionizes hydrogen. - **Balmer series** (down to $n=2$, visible): **H-alpha** = **6563 Å (red)**, H-beta = 4861 Å. These are the emission lines you see in HII regions and galaxy spectra. - **Lyman-alpha forest**: many absorption lines in a distant quasar's spectrum, each from an intergalactic HI cloud along the line of sight $\rightarrow$ maps neutral gas across cosmic time. - Energy-level math: $E_n = -13.6 \text{ eV} / n^2$. Lyman-alpha energy = $13.6(1/1^2 - 1/2^2) = 13.6 \times 0.75 = 10.2 \text{ eV}$.

Quick-recall (flash-card)

- $1+z = 1/a \cdot H^2 = (8\pi G/3)\rho - kc^2/a^2 + \Lambda c^2/3$ · acceleration needs negative pressure.
- Redshift types: **cosmological** (expansion, $1+z=1/a$), **Doppler/peculiar** (motion, $z \approx v/c$), **gravitational** ($1+GM/rc^2$); they multiply. Frames: topocentric $\rightarrow$ **heliocentric** $\rightarrow$ **CMB frame** (Sun moves $\sim 370 \text{ km/s} \rightarrow$ dipole $z \approx 0.0012$). SN Hubble diagram uses **z_{CMB} + peculiar-velocity correction**.
- $\rho \propto a^{-3(1+w)}$: matter a^{-3} ($w=0$), radiation a^{-4} ($w=1/3$), Λ constant ($w=-1$).
- $\Omega_m \approx 0.3$, $\Omega_\Lambda \approx 0.7$, flat · $q_0 = \Omega_m/2 - \Omega_\Lambda = -0.55 \rightarrow$ accelerating.
- Λ CDM = concordance. Einstein-de Sitter (matter, $a \propto t^{2/3}$). de Sitter (Λ , $a \propto e^{Ht}$).
- Dark energy: vacuum energy ($w=-1$) · **cosmological constant problem $\sim 10^{120}$** · quintessence & CPL $w(a)=w_0+w_a(1-a)$ · DESI 2024 hint of evolving DE · modified gravity · Hubble tension 67 vs 73.
- 21 cm: HI hyperfine, **$\lambda=21.1 \text{ cm}$, $\nu=1420 \text{ MHz}$** · $\nu_{\text{obs}}=1420/(1+z)$ · spin temp T_s , emission vs absorption vs CMB · Wouthuysen-Field (Lyman- α) · EDGES $z \approx 17$ · probes Dark Ages/Cosmic Dawn/EoR · LOFAR/MWA/HERA/SKA-Low.
- Hydrogen: recombination $z \approx 1100 \rightarrow$ CMB · Lyman- α 1216 Å (UV), H- α 6563 Å (red) · $E_n = -13.6/n^2 \text{ eV}$.

Likely viva Qs

1. Write the Friedmann equation and say what drives acceleration.
2. How does each component's density scale with a , and why does Λ dominate late?
3. What is the cosmological constant problem?
4. What is quintessence / evolving dark energy, and why would it matter for SN cosmology?
5. What is the 21 cm line, where does it come from, and why is it observed at low radio frequencies?
6. Difference between the 21 cm line and Lyman-alpha / H-alpha?